Exosuit Cable Stiffness Control and Co-Contraction Mechanism

`cable/cable_with_exo_springs.py`

Isaac-sim robotics notes

April 2026

1 Problem: Link 2 Strikes an Object

The cup manipulator's elbow joint q_2 is driven by antagonistic **drive cables** (green/red) via an SEA motor. During normal operation the drive cables alone govern joint motion:

$$\boldsymbol{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ \tau_2^{\text{drive}} \end{bmatrix}. \tag{1}$$

Scenario. The joint is at $q_2 = 30$ (0.524 rad) when link 2 strikes an object. The impact deflects the joint to $q_2 = 45$ — a +15 (0.262 rad) disturbance. The acceleration spike reaches $|\ddot{q}_2| = 120$ rad/s^2.

What we need: increase the joint stiffness *instantly* to resist the deflection, without changing the equilibrium position. The drive cables cannot do this alone — they are already committed to trajectory tracking.

Solution: a second set of cables — the **exosuit cables** (orange/magenta) — that provide on-demand stiffness via co-contraction. Two methods for routing these cables are compared in this document.

> **Key design question (Thrish, April 2026):** *"Are the two purple motors driving only stiffness / co-contraction, or are they driving the joint too?"*
>
> **Answer:** The exo motors drive *only stiffness*. The drive cables actuate the joint. Stiffness control is independent of and additive to joint rotation control: when the exo motors are off, the springs are slack and exert zero force regardless of q_2.

2 Physical Architecture

Cable	**Route**	**Role**	**Actuator**
Green (drive)	DrivePulley → Idler R → BigPulley → End L	Joint torque $+\tau_2$	SEA motor
Red (drive)	DrivePulley → Idler L → BigPulley → End R	Joint torque $-\tau_2$	SEA motor
Orange (exo R)	Start R → L1 Pulley R → Elbow Pulley R $\xrightarrow{\text{spring}}$ End R	Stiffness +	Exo motor R
Magenta (exo L)	Start L → L1 Pulley L → Elbow Pulley L $\xrightarrow{\text{spring}}$ End L	Stiffness +	Exo motor L

Table 1: Four cable routes on the cup manipulator.

The key design feature is the **spring element** between the elbow pulley and the link-2 anchor. When the exo motors are off, the springs sit at their rest length, so $F_{\text{exo}} = 0$ regardless of q_2. This provides *mechanical decoupling*: stiffness stays unchanged during free rotation.

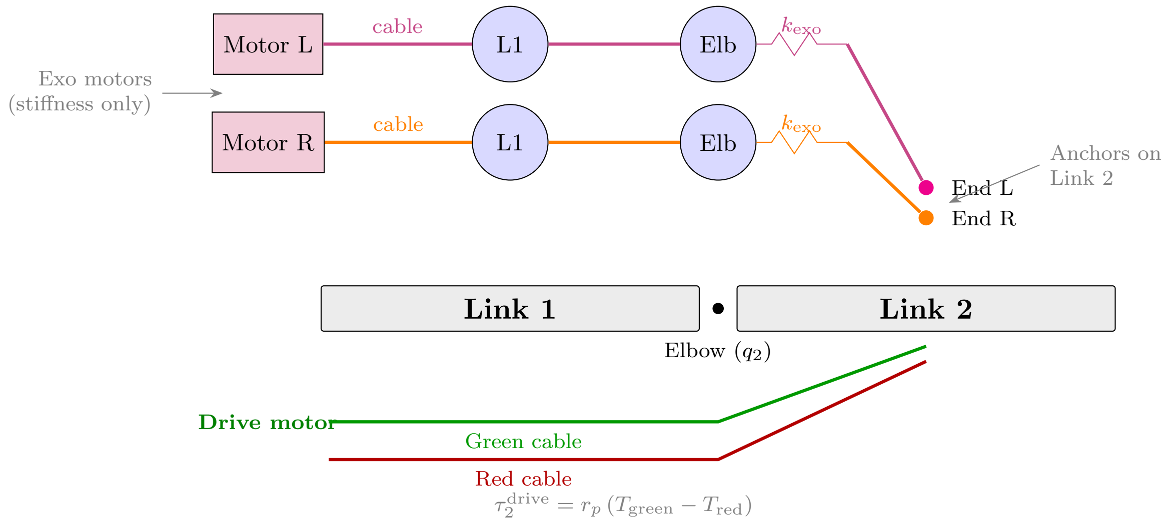

Figure 1: Dual cable architecture. Drive cables (bottom, green/red) actuate the elbow. Exo cables (top, orange/magenta) route through link-1 and elbow pulleys with a spring element before the link-2 anchor.

3 Parameters and Notation

Symbol	Description	Value	Source
Link geometry			
L_1	Link 1 length	334.8 mm	config JSON
L_2	Link 2 length	190.0 mm	config JSON
Drive cable system			
r_p	Drive pulley pitch radius	47.75 mm	HTD 5M 60T
Method A: offset pulleys			
r_{L1}	Exo L1 pulley outer radius	41.0 mm	`cable_with_exo_springs.py`
r_{elb}	Exo elbow pulley radius	32.0 mm	`cable_with_exo_springs.py`
d_{off}	Elbow pulley offset from joint	$\approx$ 103 mm	$\lvert 335 - 232 \rvert$
Method B: centred pulley (Chris)			
r_{cp}	Centred pulley radius ($= r_p$)	47.75 mm	1:1 ratio with drive
θ_m	Stepper motor angle	variable	encoder-tracking
$\Delta\theta$	Angular offset for stiffness	variable	control input
Spring parameters (shared)			
k_{exo}	Exo cable spring stiffness	200 N/m	design param
b_{exo}	Exo cable spring damping	2 N s/m	design param
Δl	Pretension pull distance	5 mm	co-contract cmd

Table 2: Notation and actual robot values from the CAD model.

4 Method A: Offset Pulleys (Current Design)

4.1 Cable geometry and the first-order approximation

Each exo cable wraps around a **Link-1 pulley** ($r_{L1} = 41$ mm) and an **Elbow pulley** ($r_{elb} = 32$ mm, centred at $x = 232$ mm on the q_1 body). The elbow *joint axis* is at $x = 335$ mm, giving an offset:

$$d_{off} = 335 - 232 = 103 \text{ mm}. \tag{2}$$

Because the pulley is *not* at the joint axis, the exact cable length from pulley exit to link-2 anchor has two q_2-dependent contributions:

$$l_{\text{cable}}^{\text{exact}}(q_2) = \underbrace{r_{\text{elb}}\,\alpha(q_2)}_{\substack{\text{cable wrap on}\\ \text{elbow pulley}}} + \underbrace{\left\| \boldsymbol{p}_{\text{exit}}(q_2) - \boldsymbol{p}_{\text{anchor}}(q_2) \right\|}_{\substack{\text{straight segment:}\\ \text{pulley exit} \to \text{link-2 anchor}}}, \tag{3}$$

where $\alpha(q_2)$ is the contact-arc angle on the pulley. The straight-segment length varies non-linearly because the link-2 anchor swings about the joint axis while the pulley centre is fixed at $d_{\text{off}} = 103$ mm away on link 1.

First-order approximation. Throughout this document we use:

$$l_{\text{cable}}(q_2) \approx r_{\text{elb}}\, q_2 \tag{4}$$

This assumes the cable departs the elbow pulley nearly tangentially and the straight-segment length changes linearly with q_2. The error is negligible for moderate angles but grows at $|q_2| > 45$. For exact tracking at large angles, a lookup table from the tangent-geometry code in `cable_with_exo_springs.py` replaces the linear approximation (see section 6.1).

4.2 Spring extension and torque

With motor cable displacements $l_{m,R}$ and $l_{m,L}$, the spring extensions are (positive q_2 = CCW):

$$\delta_R = l_{m,R} - r_{\text{elb}}\, q_2, \qquad \delta_L = l_{m,L} + r_{\text{elb}}\, q_2. \tag{5}$$

Spring force (tension only): $F_i = \max(k_{\text{exo}}\, \delta_i + b_{\text{exo}}\, \dot{\delta}_i,\ 0)$.

Net exo torque on the elbow joint:

$$\boxed{\tau_2^{\text{exo,A}} = r_{\text{elb}}\,(-F_R + F_L)} \tag{6}$$

4.3 Co-contraction: symmetric pull for pure stiffness

When a disturbance is detected ($|\ddot{q}_2| > 50$ rad/s^2), the controller captures q_2^{nom} (the pre-disturbance angle) and commands both motors symmetrically with pretension Δl:

$$l_{m,R} = r_{\text{elb}}\, q_2^{\text{nom}} + \Delta l, \qquad l_{m,L} = -r_{\text{elb}}\, q_2^{\text{nom}} + \Delta l. \tag{7}$$

Substituting into (5) with $\Delta q = q_2 - q_2^{\text{nom}}$ and ignoring damping:

$$F_R = k_{\text{exo}}\,(-r_{\text{elb}}\,\Delta q + \Delta l), \qquad F_L = k_{\text{exo}}\,(+r_{\text{elb}}\,\Delta q + \Delta l). \tag{8}$$

Net torque:

$$\tau_2^{\text{exo,A}} = r_{\text{elb}}\,(-F_R + F_L) = \boxed{2\, k_{\text{exo}}\, r_{\text{elb}}^2\, \Delta q} \tag{9}$$

The pretension Δl **cancels** — it creates stiffness only, no net torque at $q_2 = q_2^{\text{nom}}$. Effective stiffness increase:

$$k_{\text{eff}}^{\text{A}} = 2\, k_{\text{exo}}\, r_{\text{elb}}^2 = 2 \times 200 \times 0.032^2 = \boxed{0.41\ \text{N m/rad}} \tag{10}$$

4.4 Worked example: 30° impact to 45°

T$_0$: Joint at $q_2 = 30$. Exo **OFF**. $F_R = F_L = 0$.

T$_1$: Impact deflects joint to 45. Acceleration spike triggers co-contraction. Controller captures $q_2^{\text{nom}} = 30$.

T_2: Motors command via (7): $l_{m,R} = 0.0218$ m, $l_{m,L} = -0.0118$ m.
Spring extensions at $q_2 = 45$ (0.785 rad):

$$\delta_R = 0.0218 - 0.032 \times 0.785 = -0.0033 \text{ m} \;\Rightarrow\; F_R = 0 \text{ (slack)},$$
$$\delta_L = -0.0118 + 0.032 \times 0.785 = +0.0133 \text{ m} \;\Rightarrow\; F_L = 2.67 \text{ N}.$$

Restoring torque: $\tau_2^{\text{exo,A}} = 0.032 \times 2.67 = \boxed{+0.085 \text{ N m}}$

Note: Only the L cable is active — the R cable went slack because $\Delta l = 5$ mm is insufficient for a +15 deflection. If $\Delta l > r_{\text{elb}} \cdot \Delta q = 8.4$ mm, both cables stay taut and the full symmetric torque is 0.107 N m.

5 Method B: Centred Pulley + Encoder Tracking (Chris)

Chris Kalogroulis (April 2026) proposed replacing the two offset elbow pulleys with **one pulley centred on the elbow joint axis** ($r_{\text{cp}} = r_p = 47.75$ mm).

5.1 Exact cable kinematics (no approximation)

Why Method B requires no geometric approximation. Because the pulley is *coaxial* with the joint, the cable geometry is purely rotational: cable wraps/unwraps by *exactly* $r_{\text{cp}}\, q_2$ as the joint turns. The cable departure point rotates with the link-2 anchor, so the straight-segment length is **constant** for all q_2:

$$l_{\text{cable}}(q_2) = r_{\text{cp}}\, q_2 \qquad \textbf{(exact for all } q_2\textbf{)} \tag{11}$$

Contrast with Method A (eq. (3)): the offset pulley creates a four-bar-like geometry where the straight cable segment changes length nonlinearly. In Method B that complication vanishes because pulley centre ≡ joint axis.

5.2 Encoder tracking and stiffness via angular offset

Each stepper motor continuously tracks the joint encoder, feeding cable at the rate the joint moves:

$$\theta_{m,R} = q_2 + \Delta\theta, \qquad \theta_{m,L} = -q_2 + \Delta\theta. \tag{12}$$

The q_2 term keeps cables taut (zero extension, zero force during free rotation). The $\Delta\theta$ term creates symmetric pretension:

$$\delta_R = r_{\text{cp}}\, \Delta\theta, \qquad \delta_L = r_{\text{cp}}\, \Delta\theta \qquad \text{(independent of } q_2\text{)}. \tag{13}$$

Because the tracking is exact (eq. (11)), the motor compensates $r_{\text{cp}}\, q_2$ of cable per radian with *zero* geometric error, and any residual torque during free rotation is exactly zero.

5.3 Disturbance response

When a disturbance pushes the joint from q_2^{nom} to $q_2^{\text{nom}} + \Delta q$ before the stepper catches up:

$$F_R = k_{\text{exo}}\, r_{\text{cp}}\, (\Delta\theta - \Delta q), \qquad F_L = k_{\text{exo}}\, r_{\text{cp}}\, (\Delta\theta + \Delta q). \tag{14}$$

Net torque (moment arm = r_{cp}):

$$\tau_2^{\text{exo,B}} = r_{\text{cp}}\, (-F_R + F_L) = \boxed{2\, k_{\text{exo}}\, r_{\text{cp}}^2\, \Delta q} \tag{15}$$

Same structure as Method A (eq. (9)), but r_{cp} replaces r_{elb}. Effective stiffness increase:

$$k_{\text{eff}}^{\text{B}} = 2\, k_{\text{exo}}\, r_{\text{cp}}^2 = 2 \times 200 \times 0.04775^2 = \boxed{0.91 \text{ N m/rad}} \tag{16}$$

5.4 Worked example: same 30° impact to 45°

T$_0$: Joint at $q_2 = 30$. Stepper tracking encoder: $\theta_{m,R} = 0.524 + 0.15 = 0.674$ rad. Both springs pretensioned by $r_{\text{cp}} \cdot \Delta\theta = 7.16$ mm.

T$_1$: Impact deflects joint to 45. Stepper still at $\theta_{m,R} = 0.674$ rad.

T$_2$: Spring extensions:

$$\delta_R = 0.04775 \times (0.674 - 0.785) = -0.00534 \text{ m} \;\Rightarrow\; F_R = 0 \text{ (slack)},$$
$$\delta_L = 0.04775 \times (-0.524 + 0.15 + 0.785) = +0.01963 \text{ m} \;\Rightarrow\; F_L = 3.93 \text{ N}.$$

Restoring torque: $\tau_2^{\text{exo,B}} = 0.04775 \times 3.93 = \boxed{+0.188 \text{ N m}}$

2.2× more restoring torque than Method A.

T$_3$: Stepper catches up (~ 5 ms), re-centres around $q_2^{\text{nom}} = 30$ with offset $\Delta\theta$. Both cables taut; full symmetric stiffness: $\tau = 0.91 \times 0.262 = 0.239$ N m.

6 Comparison of Both Methods

Stiffness ratio. $r_{\text{cp}}/r_{\text{elb}} = 47.75/32 = 1.49\times$. Because stiffness scales as r^2:

$$\frac{k_{\text{eff}}^{\text{B}}}{k_{\text{eff}}^{\text{A}}} = \left(\frac{47.75}{32}\right)^2 = 2.23\times$$

Method B provides more than **twice** the effective stiffness for the same spring constant, simply because the lever arm is bigger.

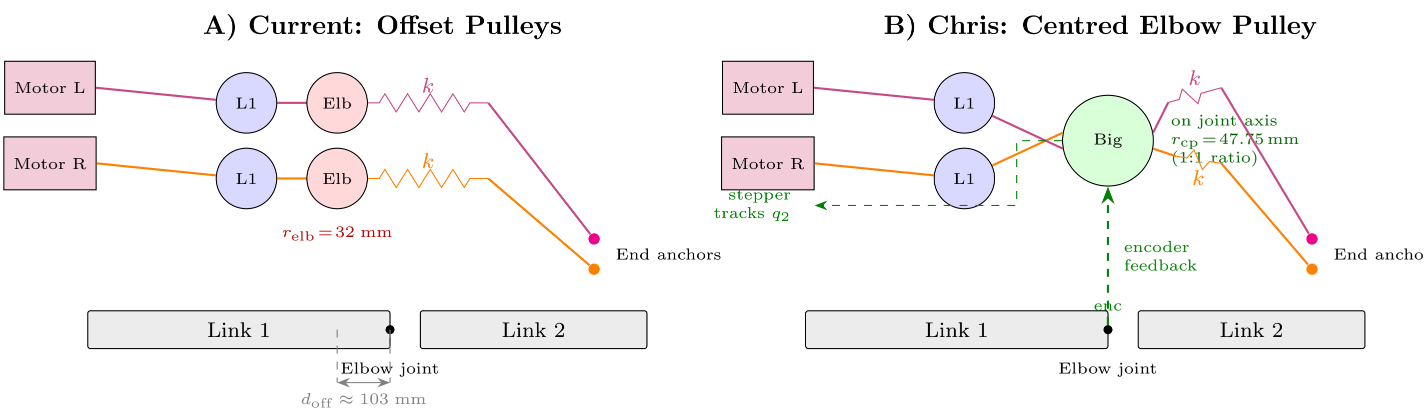

Figure 2: Routing comparison. Both methods share motor pulleys and L1 pulleys. **(A)** Two offset elbow pulleys ($r_{\text{elb}} = 32$ mm, 103 mm before joint); cable payout $\approx r_{\text{elb}}\, q_2$ (approximation). **(B)** One centred elbow pulley ($r_{\text{cp}} = 47.75$ mm, on joint axis); cable payout $= r_{\text{cp}}\, q_2$ (exact).

6.1 Method A upgrade: adding encoder tracking (A+)

Method A can also use the encoder. The motor commands become:

$$l_{m,R}(t) = r_{\text{elb}} \cdot q_2(t) + \Delta l, \qquad l_{m,L}(t) = -r_{\text{elb}} \cdot q_2(t) + \Delta l. \tag{17}$$

This keeps both cables taut at all times (no more single-cable slack), enables always-on variable stiffness, and eliminates the activation threshold. The complication is that $r_{\text{elb}} \cdot q_2$ is only an approximation (eq. (4)); exact tracking at large angles requires a lookup table from `cable_with_exo_springs.py`:

$$l_{m,R}(t) = l_{\text{cable}}\big(q_2(t)\big) + \Delta l.$$

Quantity	Method A (offset)	Method B (centred)
Pulley radius	$r_{\mathrm{elb}} = 32$ mm	$r_{\mathrm{cp}} = 47.75$ mm
Cable kinematics	$l \approx r_{\mathrm{elb}}\, q_2$ (approx.)	$l = r_{\mathrm{cp}}\, q_2$ (exact)
Effective stiffness k_{eff}	0.41 N m/rad	0.91 N m/rad
Restoring torque ($\Delta q = 15$)	0.085–0.107 N m	0.188–0.239 N m
Torque during free rotation	Small residual (geometry error)	Exactly zero
Cable slack management	Passive (springs absorb)	Active (stepper tracks encoder)
Slack risk	Low (springs provide compliance)	Depends on stepper latency

Table 3: Quantitative comparison. Both use $k_{\mathrm{exo}} = 200$ N/m.

Aspect	Method A (offset pulleys)	Method B (encoder-tracking)
Pulleys per cable	2 (L1 + offset Elbow)	2 (L1 + centred Elbow at joint)
Bandwidth requirement	Low (slow activation)	High (real-time tracking)
Single point of failure	Springs degrade gracefully	Encoder/stepper failure → slack
Prototyping effort	Springs + CubeMars	Steppers + encoder only
Large-angle accuracy	Needs lookup table	Exact ($r_{\mathrm{cp}} \cdot q_2$)

Table 4: Design trade-offs between the two routing approaches.

Bottom line. In Method B the encoder is **essential** (cables go slack without it). In Method A it is an **upgrade** that eliminates single-cable slack and enables always-on variable stiffness. The trade-off: mechanical simplicity (Method B, exact $r \cdot q_2$) versus robustness (Method A+, springs catch errors, but need a lookup table for large angles).

Prototype strategy (Chris): Build a quick first exosuit with *only* stepper motors and encoder — no springs, no CubeMars. If steppers track the encoder with < 5 ms lag, the full system is viable.

7 Dynamics and Control Summary

Full elbow dynamics with both cable systems:

$$M_{22}\,\ddot{q}_2 + c_2(\boldsymbol{q}, \dot{\boldsymbol{q}}) + g_2(\boldsymbol{q}) = \underbrace{r_p\,(T_{\mathrm{green}} - T_{\mathrm{red}})}_{\text{drive cables}} + \underbrace{r\,(-F_R + F_L)}_{\text{exo cables (when active)}} \tag{18}$$

where $r = r_{\mathrm{elb}}$ (Method A) or $r = r_{\mathrm{cp}}$ (Method B).

	A: Offset, no encoder	**A+: Offset, with encoder**	**B: Centred, with encoder**
Motor command	Fixed l_m (on trigger)	$r_{\text{elb}} \cdot q_2(t) + \Delta l$	$r_{\text{cp}} \cdot q_2(t) + r_{\text{cp}} \cdot \Delta\theta$
Cables taut?	One may go slack	Both (if Δl sufficient)	Both (if $\Delta\theta$ sufficient)
Stiffness	0–0.41 N m/rad	0.41 N m/rad	0.91 N m/rad
Activation	Threshold	Always on	Always on
Geometric accuracy	N/A	Needs lookup table	Exact

Table 5: Three operating modes. "A+" combines springs with encoder feedback.

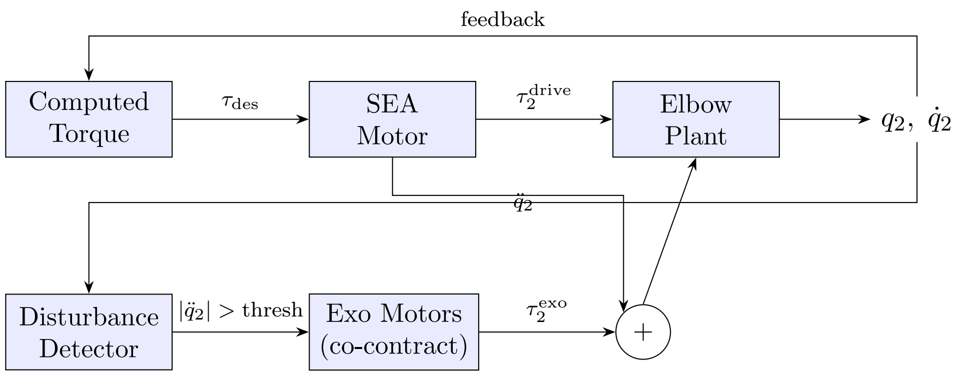

Figure 3: Control block diagram. Drive path (top) runs continuously; exo path (bottom) activates on disturbance detection.

Aspect	**Biological**	**Our mechanism**
Agonist/Antagonist	Biceps/Triceps	Drive green/red cables
Co-contraction	Both muscles activate	Both exo motors pull
Stiffness source	Muscle short-range stiffness	Exo spring k_{exo}
Activation trigger	Reflex (stretch reflex)	$|\ddot{q}_2| >$ threshold
Decoupling	Reciprocal inhibition	Spring at rest $\Rightarrow F = 0$

Table 6: Biological co-contraction analogy.